

Binary Classification Problem : w_1 vs w_2

Feature Space : $X = \mathbb{R}^2$

Training Patterns :
$$\begin{cases} \underline{x}_1 = [a \ a]^T \in w_1 \\ \underline{x}_2 = [-a \ -a]^T \in w_2 \end{cases} \quad a > 0$$

t -th training step : Decision boundary between classes w_1 and w_2 is given by:

$$g_t(x_1, x_2; \gamma_t) = x_1 + x_2 + \gamma_t \cdot a = 0$$

Consider that throughout the training process $P_t = 1, \forall t \geq 0$.

Questions: (I): Provide a graphical representation of the problem assuming that $\gamma_t = 0$.

(II): Determine the range of values for γ_t ,

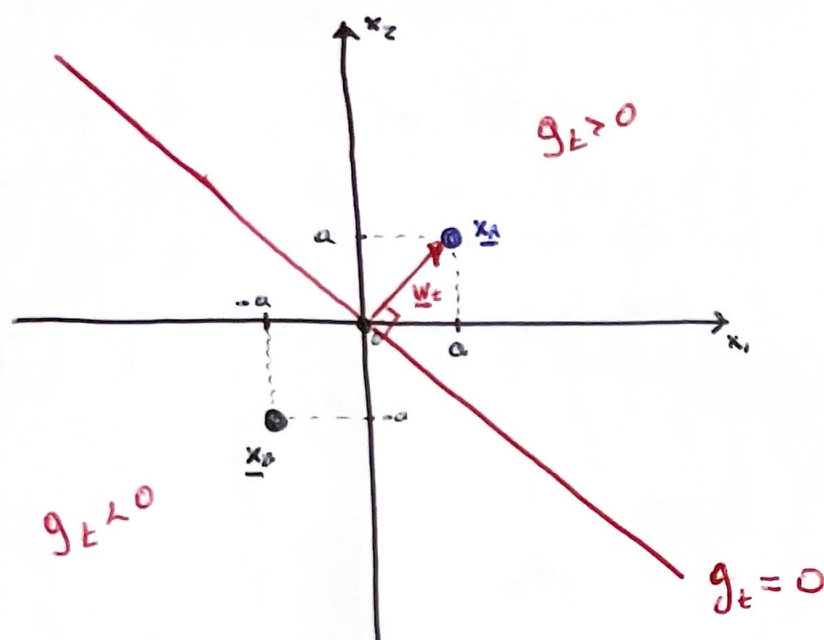
$\Delta = (\gamma_t^{\min}, \gamma_t^{\max})$ for which all patterns are classified correctly.

(III) If at the t -th training step, γ_t is out of the required range show that at the $(t+1)$ -th step the updated boundary will be given by:

$$g_{t+1}(x_1, x_2; \gamma_{t+1}) = x_1 + x_2 + \gamma_{t+1} \cdot a = 0$$

such that:

$$\gamma_{t+1} = \begin{cases} \frac{\gamma_t \cdot a + 1}{a + 1}, & \gamma_t < \gamma_t^{\min}; \\ \frac{\gamma_t \cdot a - 1}{a + 1}, & \gamma_t > \gamma_t^{\max}. \end{cases}$$

(I): Graphical Representation:

⊛ At training step t , we assume that $g_t = 0$ which yields:

$$g_t(x_1, x_2) = x_1 + x_2 = 0 \quad (1)$$

⊛ Moreover, we can deduce that $\underline{w}_t = [1 \ 1]$ and $w_0(t) = 0$. (2)

⊛ It is easy to deduce that if the value of g_t was initially equal to $g_t = 0$ both patterns $\underline{x}_A$ and $\underline{x}_B$ would be classified correctly since:

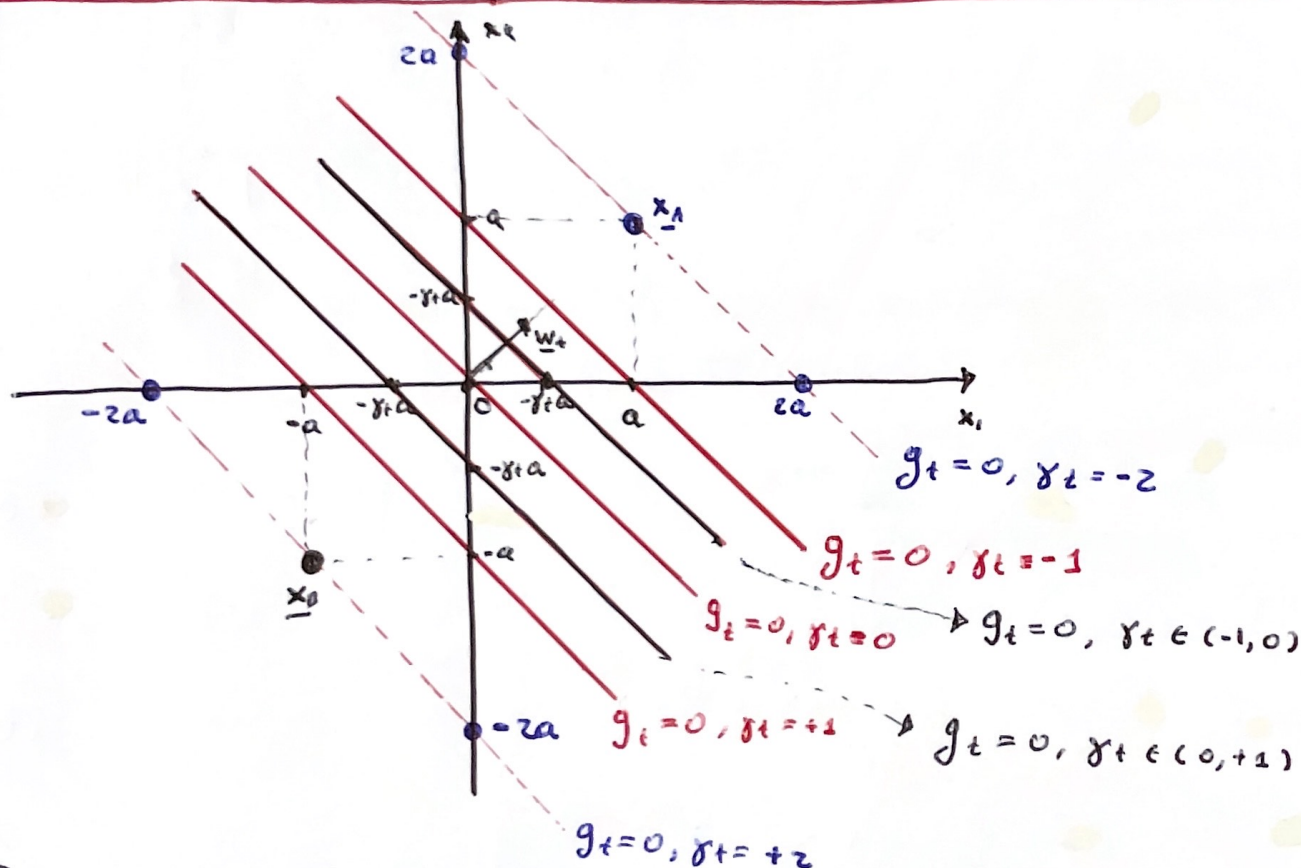
$$\begin{cases} g_t(\underline{x}_A) = g_t(a, a) = a + a = 2a > 0, \underline{x}_A \in w_1 \\ g_t(\underline{x}_B) = g_t(-a, -a) = -a + (-a) = -2a < 0, \underline{x}_B \in w_2 \end{cases} \quad (3)$$

(II): Determine A:

- ⊕ We may also deduce that in the general case where $\gamma_t \neq 0$, the discrimination line between the two classes w_1 and w_2 , $g_t(x_1, x_2) = x_1 + x_2 + \gamma_t \cdot a = 0$

(i): Hits axis x_1 for $x_2 = 0$ at: $x_1 = -a\gamma_t$ (u)

(ii): Hits axis x_2 for $x_1 = 0$ at: $x_2 = -a\gamma_t$ (s)



- ⊕ For each one of the previous cases:

$$\underline{w}_t = [1 \ 1], \quad \forall t \geq 0 \quad (6)$$

$$w_0(t) = \gamma_t \cdot a, \quad \forall t \geq 0 \quad (7)$$

⊛ It is easy to deduce that when $\gamma_t \in (-2, +2)$ all possible discrimination lines, classify all patterns correctly. [14]

$$\Delta = (-2, +2) \quad (8)$$

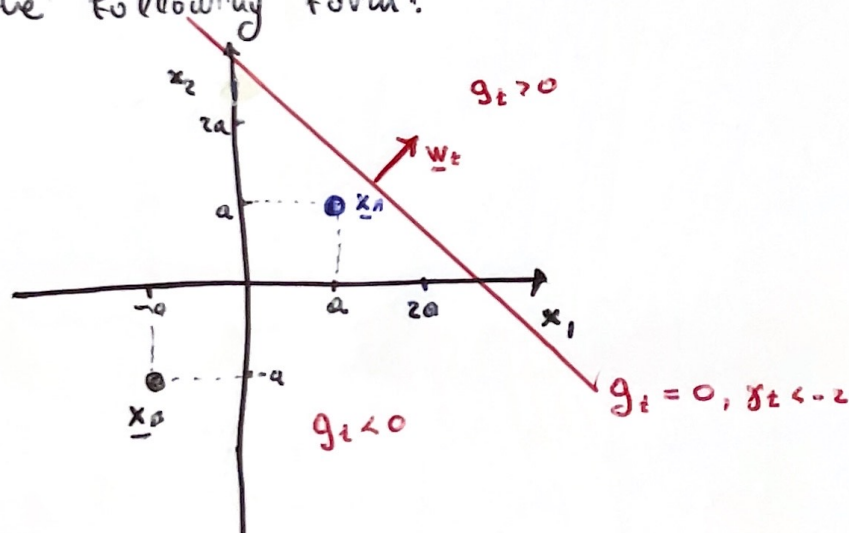
$$\gamma_t^{\min} = -2 \quad (9)$$

$$\gamma_t^{\max} = +2 \quad (10)$$

(111): Determine γ_{t+1} relative to γ_t :

Case A: $\gamma_t < \gamma_t^{\min} = -2$

For $\gamma_t < \gamma_t^{\min}$ we have that the decision boundary at time t will be of the following form:



Apparently, we have that:

$$g_t(\underline{x}_1) = g_t(a, a) = a + a + \gamma_t \cdot a = 2a + \gamma_t \cdot a < 0 \quad (11)$$

since $\gamma_t < -2$, $\underline{x}_1$ is incorrectly classified.

$$g_t(\underline{x}_2) = g_t(-a, -a) = -a - a + \gamma_t \cdot a = -2a + \gamma_t \cdot a < 0 \quad (12)$$

since $\gamma_t < -2$, $\underline{x}_2$ is correctly classified.

Since, x_A is the only pattern which is incorrectly classified,

we must form the corresponding correction vector:

$$\underline{C}_A = \delta_{x_A} \circ \underline{x}'_A \xrightarrow[\delta_{x_A} = -1]{x_A \in \omega_1} \underline{C}_A = -\underline{x}'_A \quad (13)$$

The extended weight vector at time t will be given as:

$$\underline{W}'_t = \begin{bmatrix} 1 \\ 1 \\ ax_t \end{bmatrix} \quad (14)$$

and the respective correction vector as:

$$\underline{C}_A = \begin{bmatrix} -a \\ -a \\ -1 \end{bmatrix} \quad (15)$$

We may finally write that: $\underline{W}'_{t+1} = \underline{W}'_t - p_t \cdot \underline{C}_A \Rightarrow$

$$\underline{W}'_{t+1} = \begin{bmatrix} 1 \\ 1 \\ ax_t \end{bmatrix} - \begin{bmatrix} -a \\ -a \\ -1 \end{bmatrix} = \begin{bmatrix} 1+a \\ 1+a \\ ax_{t+1} \end{bmatrix} \quad (16)$$

Ⓐ We may finally write that:

$$g_{t+1}(x_1, x_2) = (a+1)x_1 + (a+1)x_2 + (ax_{t+1}) = 0 \Leftrightarrow$$

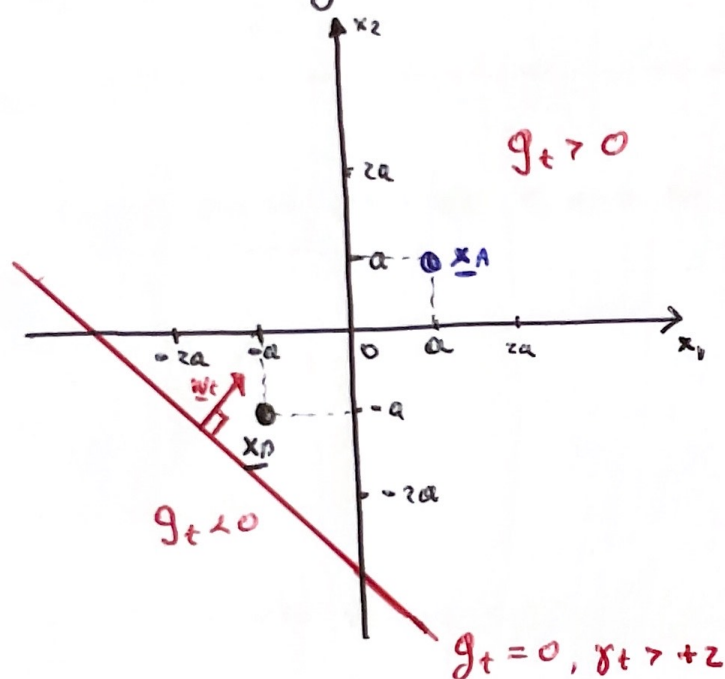
$$g_{t+1}(x_1, x_2) = x_1 + x_2 + \frac{ax_{t+1}}{a+1} = 0 \quad (17)$$

which yields

$$x_{t+1} = \frac{ax_{t+1}}{a+1} \quad (18)$$

Case B: $\gamma_t > \gamma_t^{\max} = +2$

For $\gamma_t > \gamma_t^{\max}$ we have that the decision boundary at time t will be of the following form:



Apparently, we have that:

$$g_t(\underline{x}_A) = g_t(a, a) = a + a + \gamma_t \cdot a = 2a + \gamma_t \cdot a > 0 \quad (19)$$

since $\gamma_t > +2$, $\underline{x}_A$ is correctly classified.

$$g_t(\underline{x}_B) = g_t(-a, -a) = -a - a + \gamma_t \cdot a = -2a + \gamma_t \cdot a > 0 \quad (20)$$

since $\gamma_t > +2$, $\underline{x}_B$ is incorrectly classified.

Since, $\underline{x}_0$ is the only pattern which is incorrectly classified, [17]

we must form the corresponding correction vector:

$$\underline{C}_B = \delta_{\underline{x}_0} \cdot \underline{x}_0' \xrightarrow[\delta_{\underline{x}_0} = +1]{\underline{x}_0 \in W_2} \underline{C}_B = \underline{x}_0' \quad (21)$$

The extended weight vector at time t will be given as:

$$\underline{W}_t' = \begin{bmatrix} 1 \\ 1 \\ a y_t \end{bmatrix} \quad (22)$$

and the correction vector as:

$$\underline{C}_B = \begin{bmatrix} -a \\ -a \\ 1 \end{bmatrix} \quad (23)$$

We may finally write that: $\underline{W}_{t+1}' = \underline{W}_t' - \rho_t \cdot \underline{C}_B \Rightarrow$

$$\underline{W}_{t+1}' = \begin{bmatrix} 1 \\ 1 \\ a y_t \end{bmatrix} - \begin{bmatrix} -a \\ -a \\ 1 \end{bmatrix} = \begin{bmatrix} 1+a \\ 1+a \\ a y_{t-1} \end{bmatrix} \quad (24)$$

⊛ We may finally write that:

$$g_{t+1}(x_1, x_2) = (a+1)x_1 + (a+1)x_2 + (ay_{t-1}) = 0 \Leftrightarrow$$

$$g_{t+1}(x_1, x_2) = x_1 + x_2 + \frac{ay_{t-1}}{a+1} = 0 \quad (24)$$

which yields

$$y_{t+1} = \frac{ay_{t-1}}{a+1} \quad (25)$$